Reviewer Pack — Minimal Rank of Tropical Modules over Boolean Algebras and C…

Entry 8639e90b1c59 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-05 12:36:50 UTC

Minimal Rank of Tropical Modules over Boolean Algebras and Circuit Monotone Width Inequality

Entry ID: `8639e90b1c59` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-05 12:36:50 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Tropical Modules) **Field B** (complexity object): Boolean Circuit Complexity (Circuit Monotone Width)

Statement:

> For every d-regular circuit C with n inputs, the minimal rank of a tropical module over the Boolean algebra associated with C is $O(d^{(1/2)^n}(1/3))$ and it holds that $|mtr(C)| \leq \Theta(d^{(1/2)^n}(1/3))$, where mtr(C) is the minimal rank of the tropical module over the Boolean algebra associated with C.

Rationale (proposer's reasoning):

> Tropical modules provide a geometric interpretation of boolean algebras, which could potentially expose intrinsic structure in circuit complexity. A connection between their ranks and circuit monotone width could reveal a new perspective on hardness of approximation problems or provide insights into the design of more efficient algorithms.

Taxonomy category: `TROPICAL_GEOOMETRY` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `883c9cab79152a0b`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if for all d-regular circuits C with n inputs, |mtr(C)| is within $O(d^{(1/2)^n}(1/3))$ of the actual monotone width of C, and falsified if any circuit C has |mtr(C)| more than 50% above this bound.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `tropical geometry AND circuit monotone width` - `minimal rank tropical modules AND boolean algebra circuit complexity` - `Boolean algebra associated circuits AND tropical module rank d^(1/2)n^(1/3)`

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_d_regular_circuit(n, d):
        if n % d != 0:
            return None
        circuit = [[random.randint(0, 1) for _ in range(d)] for _ in range(n)]
        return circuit

    def compute_tropical_rank(circuit):
        n = len(circuit)
        m = len(circuit[0])
        rank = 0
        for i in range(m):
            row_sums = [sum(row[i] for row in circuit) for row in circuit]
            if any(x > 0 for x in row_sums):
                rank += 1
        return rank

    def compute_monotone_width(circuit):
        n = len(circuit)
        m = len(circuit[0])
        width = 0
        for i in range(m):
            max_sum = 0
            current_sum = 0
            for j in range(n):
                if circuit[j][i] == 1:
                    current_sum += 1
                else:
```

```
                max_sum = max(max_sum, current_sum)
                current_sum = 0
            max_sum = max(max_sum, current_sum)
            width = max(width, max_sum)
        return width

    n_values = [5, 10, 15, 20, 30, 40]
    instances_tested = 0
    total_metric_value = 0.0
    conjecture_holds = True
    counterexample = ""

    for n in n_values:
        for _ in range(5):
            d = random.randint(1, min(n - 1, 2))
            circuit = generate_d_regular_circuit(n, d)
            if circuit is None:
                continue
            instances_tested += 1
            tropical_rank = compute_tropical_rank(circuit)
            monotone_width = compute_monotone_width(circuit)
            expected_bound = math.sqrt(d) * n ** (1/3)
            if abs(tropical_rank - monotone_width) > 0.5 * expected_bound:
                conjecture_holds = False
                counterexample = f"n={n}, d={d}, tropical_rank={tropical_rank}, monotone_width={mo

    return {
        "metric_name": "Monotone Width",
        "metric_value": total_metric_value / instances_tested if instances_tested > 0 else 0.0,
        "instances_tested": instances_tested,
        "n_max": max(n_values),
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19,
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, \"metric_name\": \"{result['metric_name']}\", \"metric_
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(result
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value:.6f} std={std_value:.6f} support_fraction={supp
    elif any(abs(r["metric_value"] - r["expected_bound"]) > 0.5 * r["expected_bound"] for r in res
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)


```
 463, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 26, "n_max": 40,
TRIAL: {"seed": 503, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
TRIAL: {"seed": 547, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
TRIAL: {"seed": 593, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
TRIAL: {"seed": 631, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
TRIAL: {"seed": 677, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
TRIAL: {"seed": 727, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
TRIAL: {"seed": 773, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
TRIAL: {"seed": 821, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
TRIAL: {"seed": 877, "metric_name": "Monotone Width", "metric_value": 0.000000, "instances_tested": 2
```


8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> Safety rail: critic_challenge_falsified | original: The test results show that for n=40 and d=2, the tropical rank exceeds the conjectured bound by more than 50%, with a monotone width of 5 when the exp | next: Investigate circuits that lead to tropical ranks exceeding the conjectured bounds and explore potential reasons for this discrepancy.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13334
2	preregistration	ollama_remote	glm4:latest	0	0	9830
3	novelty	ollama_remote	glm4:latest	0	0	10701
4	novelty	ollama_remote	glm4:latest	0	0	16523
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16910
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11952
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12907
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12094
9	judge	ollama_remote	glm4:latest	0	0	12827

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 117079 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/8639e90b1c59.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/8639e90b1c59.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/8639e90b1c59.tar.gz` (if generated)